M4C 2025 - Assignment 1

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Problem 1

(a) encrypted text: XEFVRXEFWPDJAFPCCBHBRVMDUJ
decryption process: To get back the plain text, you need to use the same 'lenghtened' key but now you subtract the number of the letter in the key from the corresponding number of the letter in the encrypted text.

(b) Let $p_1, p_2, \ldots, p_n$ represent letters in P, $k_1, k_2, \ldots, k_n$ letters in 'lenghtened' K and $c_1, c_2, \ldots, c_n$ letters in C.
Encryption: $c_i = (p_i + k_i) \mod 26$
Decryption: $p_i = (c_i - k_i) \mod 26$

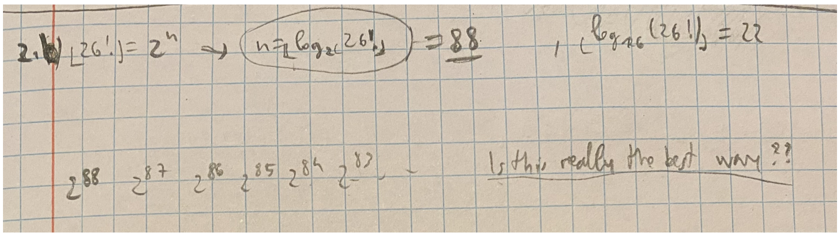
(c) No matter which 2 out of the 3 items are captured by the adversary, since you always catch at least a key or plain text. Knowing at least one of them can lead to knowing the other.
Captured P and K: encryption
Captured P and C: you can write down the differences between letters and derive the lenghtened key. From there you just take the first iteration.
Captured K and C: decryption

(d) We can use this table for both encrypting and decypting process. We can imagine that the first row (or symmetrically the first column) represents the key and the furst column (or symmetrically the first row) the plain text. For every letter in the plain text we can lookup its corresponding letter in the cipher text by taking the letter in the row of the letter (in plain text) and column of the letter from the key (in corresponding place). We can also use this table for decryption process by again imagining the first row/column as plain text and first column/row as key. Since we only know the cipher text and key this time, for every letter in cipher text we need to first find the letter that corresponds to it in key. We then look into that (from the key) letter's row until we find the letter from the cipher text that responds to it. The column in which this letter is situated belings to the letter in the plain text.

(e) for this exercise please look at the second pdf!

Problem 2

(a) A number ends in exactly 1 zero if it is a multiple of 10 and not a multiple of 100. If a number ends in exactly i zeros, it is a multiple of 10^i and not a multiple of 10^{i+1}. I don't know why this number ends in exactly 6 zeros.

(b) 

I tried to do it by hand but doing 3 * 88 calculations didn't sound appealing or worth it. Therefore, I suspected that I must be doing something wrong so if that's the case, please instruct me!

(c) The easiest way to do this would be just taking every fourth element of question (b) but since I didn't complete it and doing 22 * 3 calculations still doesn't sound that efficient to me, I'm leaving this question unanswered :)

Problem 3

(a) The most important parts of enigma are: battery, key switches, plugboard, rotors and light bulbs. Encryption process starts with pressing down the switch of the corresponding letter in the plain text. Electricity runs through this key switch and to the plugboard, where the letter can already be changed if there is a cable attached to it. Electricity in that case runs through the cable to the other letter and to the 3 rotors. Each of them will definitely change the letter twice. The letter will also be changed once at the reflector. Electricity then runs back to the plugboard to check if the newest letter has a cable connected to it. From there it goes to the corresponding key switch and to the light bulb (where it turns it on). The circut needs to be connected so it runs back to the battery.

(b) First, we need to choose 3 out of 5 rotors in total: $\binom{5}{3}$. We then need to learn their order: 3!. There are 26 letters on each one and we need to choose the starting position for each one: 26^3. The last thing we need to learn is the positions of the cables. For this part I came to the conclusion in a different way than James Grimes. My reasoning follows: First you need to choose a pair out of 26 letters: $\binom{26}{2}$. After that you again need to choose a pair, but only out of 24 letters, since the letters cannot be connected to 2 different letters: $\binom{24}{2}$. We follow the same logic 8 more times (10 in total) to get equation: $\binom{26}{2}\binom{24}{2}\binom{22}{2}\binom{20}{2}\binom{18}{2}\binom{16}{2}\binom{14}{2}\binom{12}{2}\binom{10}{2}\binom{8}{2}$. Simplifying this equation gives us: $26!/(2^{10} * 6!)$. The last thing we need to take into account is shuffling: 1/10!. This results in the wanted formula.

(c) If we neglect the first part of the formula, there is exaclty 1 combination for the plugboard if we don't use any plugs. If we use exactly 1 plug, we need to choose 2 out of 26 letters. If we use exactly 2 plugs, we first need to choose 2 out of 26 and then 2 out of 24 letters. The order of the two pairs also does not matter, resulting in the following formula:

$$1 + \binom{26}{2} + \frac{\binom{26}{2}\binom{24}{2}}{2!}$$

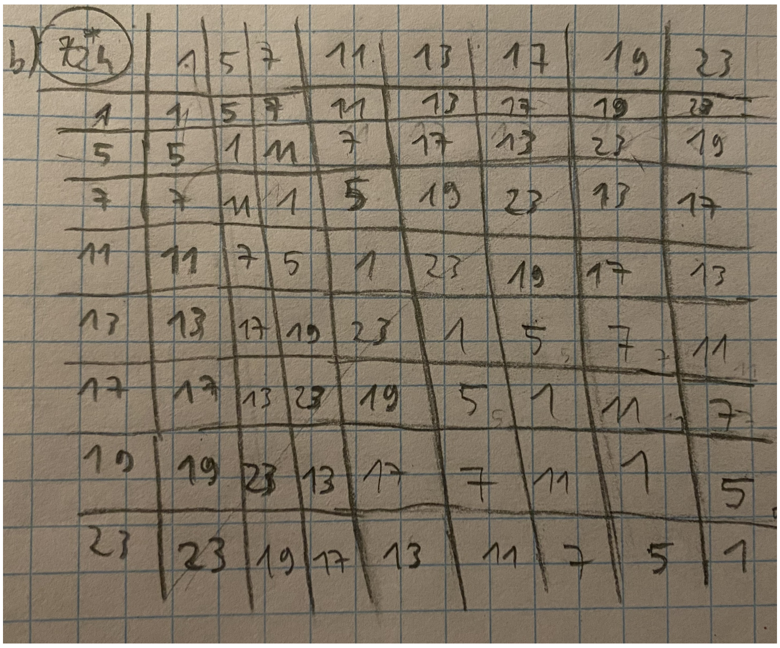
(d) The total number of unrestricted mappings for 4 letters is just permutations of 4 elements, so $4! = 24$. The number of mappings where no letter mapps to itself, however, is only 9.

Because the letter would never map onto itself, the cipher became easier to crack. Even in our example with only 4 elements there are 24 possible combinations where we aren't even considering the rotor contribution. If a letter never maps onto itself, the whole equation becomes smaller. However, the biggest implication this provided was that if you knew of a word or two that would most definitely be included in the cipher every day, you could try to look for that sequence of letters and only try the combinations where the letters wouldn't overlap. This reduces the amount of possibilities greatly.

Problem 4

(a) 1, 5, 7, 11, 13, 17, 19, 23 (so basically any number lower than 24 that is composed of only primes that don't include 2 and 3, since $24 = 2^3 3$

Because we there are 8 numbers, lower than 24 that are its coprimes, we have: $\varphi(24) = 8$)

(b) 

b) $\mathbb{Z}_{24}^*$	1	5	7	11	13	17	19	23
1	1	5	7	11	13	17	19	23
5	5	1	11	7	17	13	23	19
7	7	11	1	5	19	23	13	17
11	11	7	5	1	23	19	17	13
13	13	17	19	23	1	5	7	11
17	17	13	23	19	5	1	11	7
19	19	23	13	17	7	11	1	5
23	23	19	17	13	11	7	5	1

Interesting fact I noticed: Not only is the matrix symmetrical over the $x = -y$ diagonal (sorry I don't know how to describe it better), it is also symmetrical over the $x = y$ diagonal.

(c) An inverse of a unit is the unit itself. We can learn this by looking at the diagonal elements of the table in (b).

(d) The sum of all units is 0 and the product of all units is 1. It would seem that we get precisely the units of addition and multiplication in $\mathbb{Z}_{24}$. The reasoning in is under the next question.

(e) Firstly, we have: $\varphi(11) = 10$, therefore $\varphi(11)/11 = 10/11 \approx 1$, because 11 is prime. The sum of all elements is again 0 and the product is 10.

Next, we have: $\varphi(15) = 8$, therefore $\varphi(15)/15 = 8/15 \approx 1/2$. The sum of all elements is again 0 and the product is again 1.

It seems that the sum of elements is always 0 no matter the amount of elements in our space. After some thinking I have noticed that there is always an even amount of elements in our space that are each others' inverses in both addition and multiplication spaces. In multiplication space, the element is its own inverse, whereas in the addition space, the inverse of the j-th element is element (n - j) where n is the modulo. Since we have an even amount of elements in our space and (for some reason***) for every element in the space, there exists its inverse (by addition). Example: in n = 9, the elements in our space are: 1, 2, 4, 5, 7, 8. So when adding them all up, we can add 1 and 8 to form 9 (= 0), 2 and 7 to form 9 (= 0) and 4 and 5 to again form 9. I don't know how to prove that this is true for all n, if it even is.

The product, however, is dependent on the number of elements in our space. When we had 8 elements, the product was 1 and when we had 10 elements, the product was 10. Upon further thinking, I noticed that if the numeber of elements in our space is a power of 2, the product is always 1. Furthermore, I also tried what happend when the number of elements is 6 (I toon n = 9) and noticed that the product was 8, which was (n - 1). Since there is always an even number of elements in our space, I can only suspect that this holds for every other n, which space doesn't have a power of 2 elements.

***Further thinking: Maybe this is true because if we subtract n from (n - j), we get -j, which is not devisible by any factor from n, since j and n are coprime.